Computer Graphics
Unit 1: Introduction and Graphics Hardware

Classroom notes that go from basic ideas to exam-ready depth, with diagrams, worked numbers, comparison tables and practice questions.

B.Tech CSE, 3rd Year | Subject: Computer Graphics | Unit 1 | Lecture notes with practice set

A note before you start

The headings follow the standard textbook order for this unit (for example, the order used in Hearn and Baker's *Computer Graphics*). Universities differ in weightage and in the exact wording of the syllabus, so match these notes against your own syllabus document and past papers. Numbers marked *assumed* are teaching values, not hardware standards.

Learning outcomes

After this unit you should be able to:

- **Define** computer graphics and **list** its major application areas.
- **Explain** how a CRT, a raster-scan system and a random-scan system work.
- **Compare** raster and random scan, and CRT with flat-panel displays.
- **Calculate** frame-buffer size and basic refresh timings.
- **Select** suitable display and input devices for a given application.

1. What is Computer Graphics?

Start here: the big picture

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Definition: Computer Graphics

Computer graphics is the field that deals with creating, storing, manipulating and displaying pictures using a computer. A picture can be a drawing, a chart, a photograph-like scene or an animation.

Every graphics program does the same basic job: it takes **data** (numbers, shapes, a 3D model) and turns it into **pixels** on a screen. Everything in this unit is about the hardware that holds and shows those pixels.

Real-world example

When you open Google Maps on your phone, the app holds road data (coordinates of lines and areas). The graphics system turns that data into coloured pixels, and your touch on the screen is the input that changes the picture. Data, picture and interaction: that is computer graphics in one example.

Types of computer graphics

- **Passive (non-interactive):** the user only watches. Example: a movie or a pre-rendered animation.
- **Interactive:** the user changes the picture in real time. Example: a video game or a CAD drawing.
- **2D and 3D:** 2D graphics use x and y only (charts, icons). 3D graphics also use depth z (car models, games).

Graphics vs image processing vs computer vision

Field	Input	Output	Example
Computer graphics	Data or a model	An image	Rendering a 3D car from a design file
Image processing	An image	A modified image	Sharpening a blurry photo
Computer vision	An image	Information or decisions	Detecting faces in a photo

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Remember

Graphics goes data to image. Computer vision goes image to data. They are roughly opposite directions.

2. Application Areas of Computer Graphics

Where do we use it?

Area	What graphics does there	Real-world example
Computer-aided design (CAD/CAM)	Draws and tests engineering and architectural designs on screen before anything is built	AutoCAD for building plans; SolidWorks for machine parts
Presentation graphics	Turns numbers into charts and graphs for reports	Bar and pie charts in Excel or PowerPoint; business dashboards
Computer art	Lets artists paint, sculpt and animate digitally	Photoshop, Blender, digital illustration on a tablet
Entertainment	Creates film effects, animated movies and games	Animated films, mobile and PC games
Education and training	Simulates situations that are costly or risky to practise for real	Flight and driving simulators; virtual science labs
Visualization	Shows complex scientific or business data as pictures	Weather maps, 3D models of molecules, CT and MRI volume views
Image processing	Improves or analyses existing pictures	Photo filters; enhancing satellite images
Graphical user interfaces (GUI)	Gives windows, icons, menus and buttons instead of typed commands	Windows, Android and iOS screens
Virtual and augmented reality	Places the user inside, or overlays, a computer-made world	VR headsets; AR apps that place objects in your room via the camera

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Area	What graphics does there	Real-world example
Maps and GIS	Draws geographic data in layers	Navigation apps; city planning maps

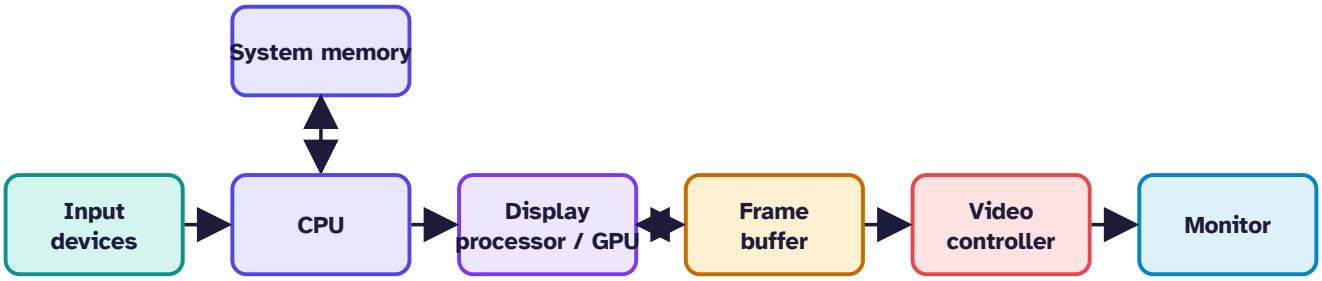
Important point for exams

Do not just list names. For each area, add one line on what graphics does and one example. Four well-explained areas score better than ten bare names.

3. Overview of Graphics Systems

The parts of the machine

A graphics system is a computer system with extra support for making and showing pictures. It has **hardware** (processors, memory, display, input devices) and **software** (the application program and a graphics library such as OpenGL that talks to the hardware).



Picture data flows left to right. The monitor shows what the frame buffer holds.

Figure 1: Block diagram of a general graphics system

Component	Job in simple words
Input devices	Let the user give positions, choices and commands (keyboard, mouse, tablet).
CPU and system memory	Run the application and store the picture description.
Display processor / GPU	Converts shapes into pixel values so the CPU is free for other work.

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Component	Job in simple words
Frame buffer	Memory that stores the colour or intensity of every pixel on screen.
Video controller	Reads the frame buffer again and again and feeds the monitor.
Monitor	Turns the signal into visible light.

Definition: Frame buffer (refresh buffer)

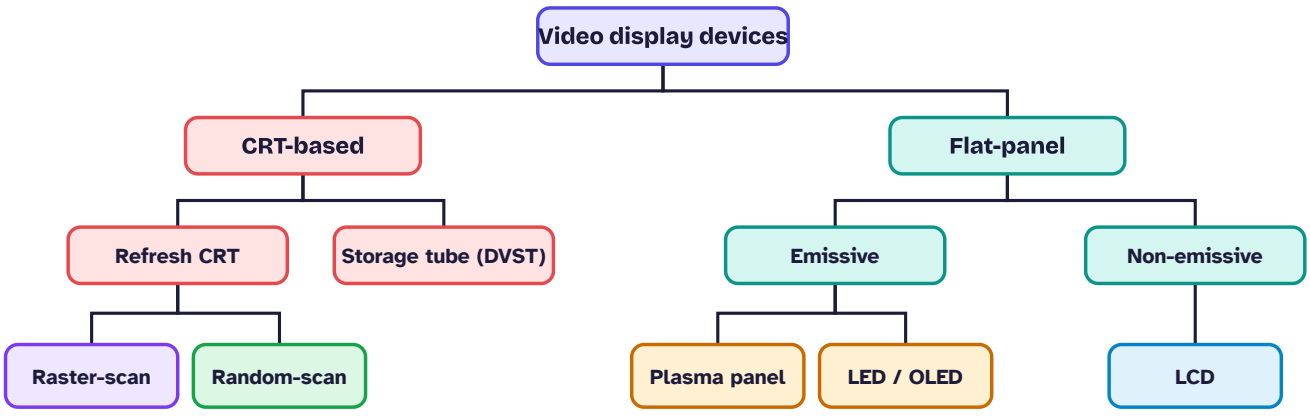
A frame buffer is a block of memory that holds the intensity or colour value of every point (pixel) on the screen for one frame.

Going deeper

Modern PCs and phones use flat-panel displays, but the same idea survives: the GPU draws into video memory, and a display controller scans that memory out to the panel at a fixed rate. The older CRT hardware in this unit is how that idea was first built, and it is still what exams ask about.

4. Video Display Devices

How the picture becomes light



Emissive panels make their own light. Non-emissive panels modulate light from a backlight or from the room.

Figure 2: Classification of video display devices

4.1 Refresh cathode-ray tube (CRT)

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The CRT was the main display in graphics for decades. Understanding it makes raster and random scan easy.

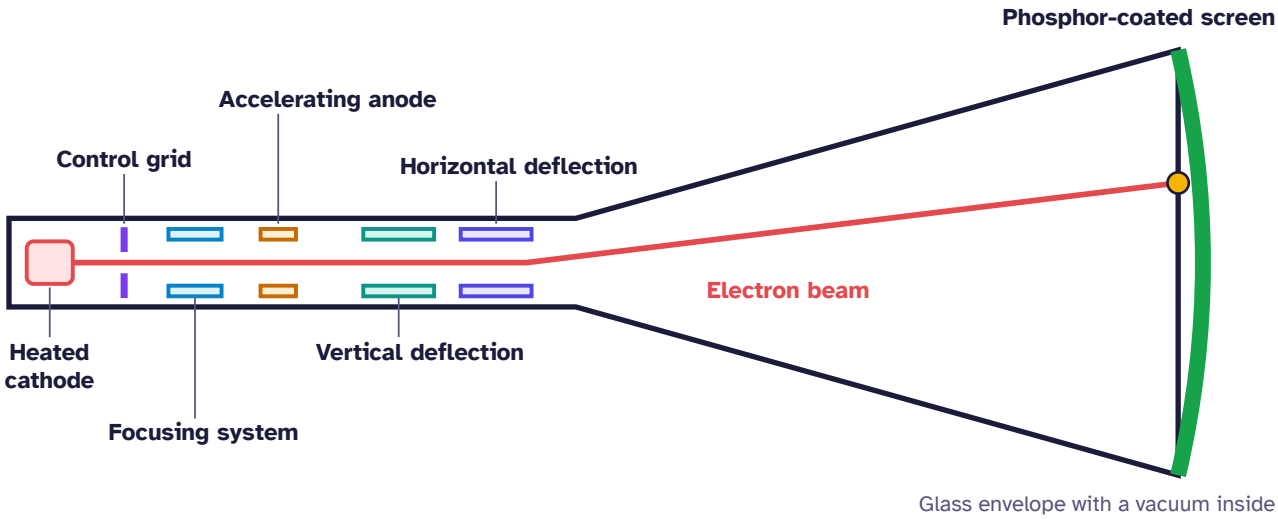


Figure 3: Cross-section of a refresh CRT (deflection can be done with plates or with magnetic coils)

How it works, step by step

1. The **heated cathode** releases electrons.
2. The **control grid** controls how many electrons pass. Fewer electrons means a dimmer spot, more means a brighter spot.
3. The **focusing system** squeezes the electrons into a thin beam so the spot is small and sharp.
4. The **accelerating anode** gives the electrons high speed.
5. The **deflection system** (horizontal and vertical) steers the beam to the required screen position.
6. When the beam hits the **phosphor coating**, that spot glows. The glow fades quickly, so the beam must redraw the picture again and again. That is why it is called a **refresh** CRT.

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Key terms for CRT

- **Persistence:** the time a phosphor keeps glowing after the beam leaves. It is usually measured as the time for the light to decay to one-tenth of its original intensity.
- **Refresh rate:** how many times per second the picture is redrawn. Rates of about 60 Hz or higher are used to avoid visible flicker.
- **Resolution:** the number of distinct points (pixels) that can be displayed without overlapping, for example 1920 × 1080.
- **Aspect ratio:** the ratio of screen width to height, such as 4:3 or 16:9. (Some textbooks define it as the ratio of vertical to horizontal points needed to draw lines of equal length. Use the definition from your textbook.)
- **Retrace:** the quick movement of the beam back to the start of a line or to the top of the screen, with the beam switched off.

Why persistence and refresh rate are linked

Phosphor with **short persistence** fades fast, so it needs a **higher refresh rate**. Phosphor with **long persistence** can be refreshed less often, but moving objects leave smeary trails. Animation therefore prefers short persistence.

Going deeper

The brightness across a beam spot is not flat. It is brightest at the centre and falls off towards the edge, roughly like a Gaussian curve. Two neighbouring spots are considered "separate" only if they are far enough apart that their glows do not merge. This is why resolution depends on spot size, not only on how many addresses the electronics can generate.

4.2 Colour CRT monitors

Colour is created by mixing red, green and blue light. There are two classic methods.

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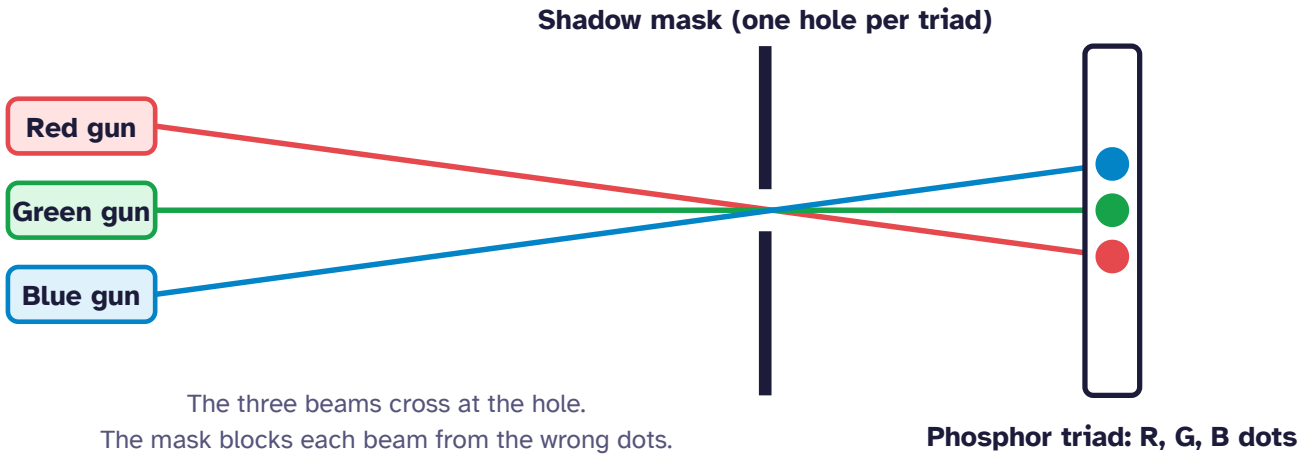


Figure 4: Shadow-mask colour CRT (simplified side view)

Point	Beam-penetration method	Shadow-mask method
Idea	Screen has two phosphor layers (red and green). Colour depends on how deep the electrons reach.	Each pixel is a triad of red, green and blue dots hit by three separate beams.
Control of colour	Change the beam speed (anode voltage).	Change the intensity of each of the three beams.
Number of colours	Very few: red, green, orange, yellow.	A very large range, since each beam has many intensity levels.
Picture quality	Lower.	Higher.
Typically used with	Random-scan displays.	Raster-scan displays (TVs and monitors).
Cost	Cheaper.	More expensive.

4.3 Direct-view storage tube (DVST)

A DVST keeps the picture **stored on the tube itself** (on a fine mesh behind the screen), so there is no refresh and no flicker even for very complex line drawings. The disadvantages are that individual parts of the picture **cannot be erased** (the whole screen must be wiped), so interactive editing and animation are not possible, and colour is not available. DVSTs are of historical interest today, but they are still asked as a "short note".

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4.4 Flat-panel displays

Flat panels are thin, light and use less power than a CRT. They are of two kinds.

- **Emissive displays** convert electrical energy into light themselves. Examples: plasma panels, thin-film electroluminescent displays and LED/OLED displays.
- **Non-emissive displays** use ambient light or a backlight and control how much of it passes through. Example: liquid crystal displays (LCD).

Feature	CRT	Plasma	LCD (LED-backlit)	OLED
Principle	Electron beam hits phosphor	Gas discharge lights phosphor cells	Liquid crystals block or pass backlight	Organic material emits light when current flows
Own light?	Yes (emissive)	Yes (emissive)	No (non-emissive)	Yes (emissive)
Thickness and weight	Bulky, heavy	Thin	Very thin	Very thin
Power use	High	High	Low	Low to medium
Typical use today	Mostly obsolete	Large TVs (largely replaced)	Monitors, laptops, TVs	Phones, premium TVs

Going deeper: how an LCD pixel works

Light from the backlight passes through a polarizing filter, then through a layer of liquid crystal, then through a second polarizer turned at right angles. With no voltage the crystal twists the light's polarization so it passes the second filter. When voltage is applied the crystal untwists and the light is blocked. In **active-matrix** panels a transistor (TFT) sits at every pixel to hold this voltage precisely, giving sharper and faster pictures than **passive-matrix** panels.

4.5 Three-dimensional viewing devices and VR

To create the feeling of depth, a system must show each eye a slightly different image. Stereoscopic displays do this with special glasses or with separate screens. **Head-mounted**

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displays (HMD) put a small screen in front of each eye and track the head movement so the view changes as the user turns. This is the basis of virtual-reality headsets.

5. Raster-Scan Systems

The display used by almost every screen today

Definition: Raster scan

In a raster-scan display the electron beam sweeps across the screen one row (scan line) at a time, from top to bottom. The picture is stored as a grid of pixel values in a frame buffer, and the beam intensity is set for every pixel as it passes.

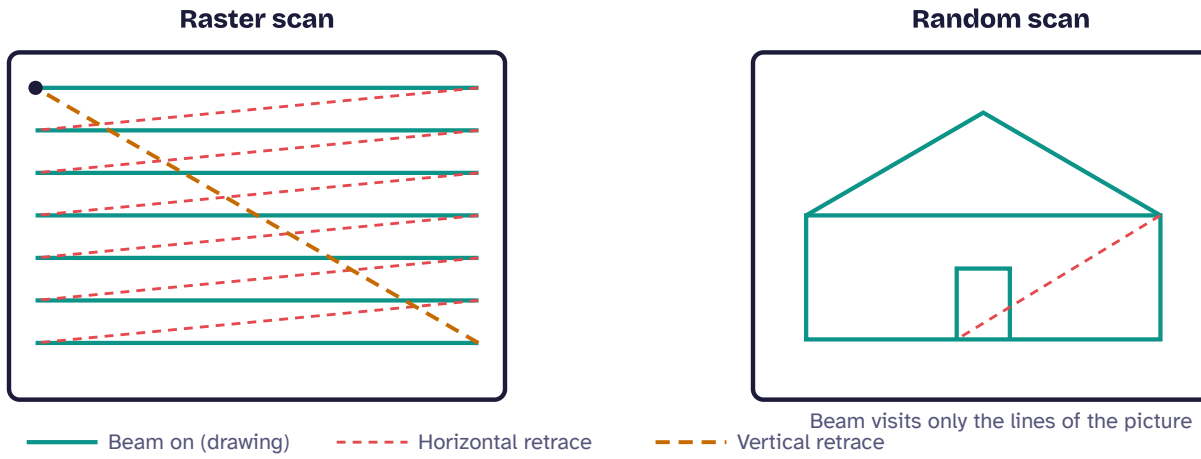


Figure 5: Raster scan sweeps every row. Random scan jumps only to the lines that exist.

Important terms

- **Scan line:** one horizontal row of pixels.
- **Horizontal retrace:** the beam returns from the end of one line to the start of the next, with the beam blanked (off).
- **Vertical retrace:** after the last line, the beam returns from the bottom corner to the top-left corner, again with the beam off.
- **Bitmap:** a frame buffer with **1 bit per pixel** (each pixel just on or off).
- **Pixmap:** a frame buffer with **more than 1 bit per pixel** (grey levels or colours).
- **Interlacing:** the screen is refreshed in two passes. First all odd-numbered lines, then all even-numbered lines. Each pass is called a **field** and two fields make one frame. This reduces visible flicker when the full-frame rate is low.

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Architecture of a raster-scan system

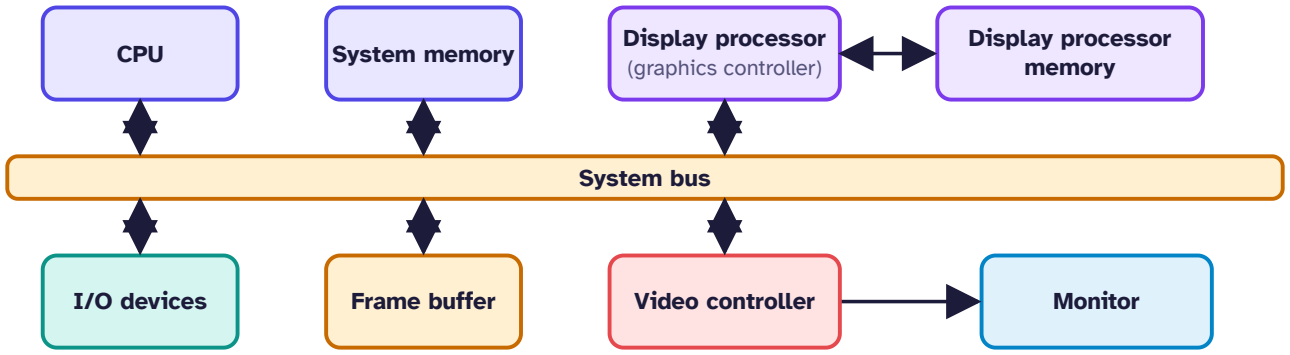


Figure 6: Raster-scan system with a separate display processor (the frame buffer may also sit inside system memory in simpler systems)

Role of each new part

- **Display processor (graphics controller):** takes shapes such as lines, circles and filled areas from the CPU and converts them into pixel values in the frame buffer. This conversion is called **scan conversion**. It frees the CPU from drawing every pixel.
- **Video controller:** keeps reading the frame buffer in scan-line order, at the refresh rate, and controls the beam.

How the video controller works (simple version)

1. It keeps two counters, `x` (column) and `y` (row), set to the top-left pixel.
2. It reads the pixel value at (x, y) from the frame buffer and uses it to set the beam intensity.
3. It increases `x` to move to the next pixel. At the end of a row, `x` returns to `0` and the next row starts (horizontal retrace).
4. After the last row, it returns to the first row (vertical retrace) and the cycle repeats, `60` or more times per second.

The frame buffer holds numbers, not shapes

If a program draws a red line, the frame buffer simply contains red values at the pixels along that line. It does not remember that a "line" exists. To move the line, the program must clear those pixels and write new ones.

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Going deeper

- **Jagged edges (aliasing):** since pixels are a grid, slanted lines look like staircases. Anti-aliasing reduces this.
- **Colour lookup table:** instead of storing a full colour per pixel, the frame buffer may store a small index (for example 8 bits) that points to a table of actual colours. This saves memory.
- **Double buffering:** the program draws the next frame in a hidden buffer while the screen shows the current one, then swaps. This prevents half-drawn frames in animation.

Practical examples: numbers you may be asked to calculate

Formula

Frame-buffer size (bits) = horizontal pixels × vertical pixels × bits per pixel.
Divide by 8 for bytes, then by 1024 for KB, and again by 1024 for MB.

Example 1: Frame buffer size

Question: Find the frame-buffer size for a 1024 × 768 display at (a) 8 bits per pixel and (b) 24 bits per pixel.

Pixels = 1024 × 768 = 786,432.
(a) 8 bpp: 786,432 bytes = 768 KB.
(b) 24 bpp = 3 bytes per pixel: 786,432 × 3 = 2,359,296 bytes = 2304 KB = 2.25 MB.

Example 2: Colours from bit depth

Number of colours = $2^{\text{bits per pixel}}$. So 1 bit gives 2 colours (a bitmap), 8 bits give 256, 12 bits give 4096 and 24 bits give 16,777,216.

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Example 3: Time available per pixel

Question: A 1280 × 1024 raster display is refreshed at 60 Hz. Ignoring retrace time, how long can the system take per pixel?

Pixels per frame = 1280 × 1024 = 1,310,720.
Pixels per second = 1,310,720 × 60 = 78,643,200.
Time per pixel = 1 / 78,643,200 ≈ 12.7 nanoseconds. Real systems have even less because retrace time is lost.

Example 4: Interlacing

If a display shows 60 fields per second with interlacing, a complete frame needs two fields, so the full picture is updated 30 times per second. Each field carries only half of the scan lines (for a 1080-line picture, 540 lines per field).

6. Random-Scan Systems

Also called vector, stroke-writing or calligraphic displays

Definition: Random scan

In a random-scan display the electron beam is directed only to the parts of the screen where a picture exists, and it draws the picture one line segment at a time, in any order. The picture is stored as a set of drawing commands, not as pixels.

The stored command list is called the **display file**, **display list** or **refresh display file**. A typical command says "draw a line from (x1, y1) to (x2, y2)". The **display processor** (also called the display processing unit or graphics controller) reads this list and draws each line, then goes back to the top and repeats, many times per second.

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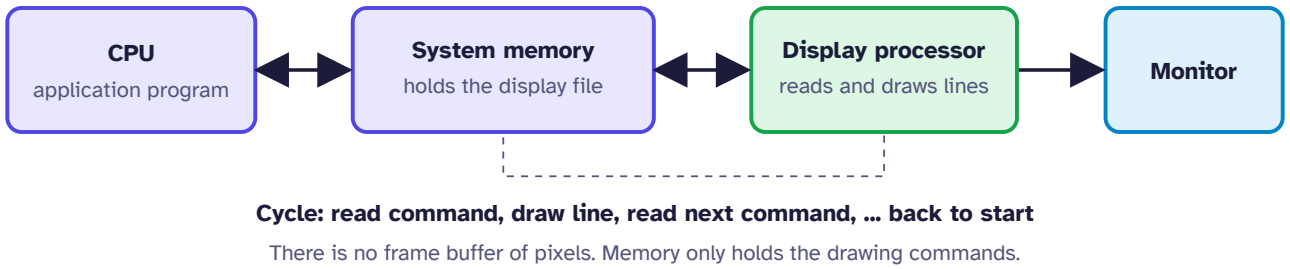


Figure 7: Random-scan system. The display processor repeatedly cycles through the display file.

Refresh depends on content

In random scan, the refresh time depends on how many lines are in the picture. Too many lines and the beam cannot redraw them all in time, so the screen **flickers**.

Example 5: How many lines can be refreshed? (assumed values)

Suppose a random-scan display must refresh at 60 Hz and each line takes about 20 microseconds to draw.
Time per refresh = 1/60 s ≈ 16,667 microseconds.
Maximum lines ≈ 16,667 / 20 ≈ 833 lines. A drawing with 2000 such lines would refresh at only about 25 Hz, which will flicker.

Example 6: Memory comparison

A drawing of 2000 lines, each stored as four coordinates of 2 bytes, needs 2000 × 8 = 16,000 bytes (about 15.6 KB). A 1024 × 768 raster frame buffer at 8 bpp needs 768 KB regardless of how simple the picture is. Random scan uses far less memory for simple line drawings.

Raster scan vs random scan

Feature	Raster scan	Random scan
Also called	Pixel-based display	Vector, stroke-writing, calligraphic display

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Feature	Raster scan	Random scan
How the beam moves	Sweeps every line, top to bottom	Goes only where the picture has lines
Picture stored as	Pixel values in a frame buffer	Line-drawing commands in a display file
Memory needed	Depends on screen resolution and colour depth	Depends on the number of lines
Refresh time	Constant, independent of picture	Varies with number of lines
Line quality	Slanted lines can look jagged (aliasing)	Smooth lines
Filled areas and shading	Handled well	Poor
Realistic scenes and photos	Yes	No
Colour	Millions of colours possible	Very limited (for example, beam penetration)
Best for	TV, games, general computing, photographs	Wireframe line drawings, plotters
Today	Used by nearly every screen	Mostly historical; the idea lives on in vector graphics software

Exam tip

A 10-mark answer on this topic needs: both definitions, both architecture diagrams, a comparison table of at least 6 points, and one line on why raster scan won (cheap memory, shading, colour).

7. Graphics Monitors and Workstations

Putting hardware together for real work

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Graphics monitors

A graphics monitor is a display with enough **resolution, colour depth and refresh rate** to show detailed pictures. Typical specifications to know:

Specification	Meaning
Screen size	Measured along the diagonal (for example, 24 inch).
Resolution	Pixels across × pixels down (for example, 1920 × 1080).
Colour depth	Bits per pixel; more bits mean more colours.
Refresh rate	Redraws per second (Hz). Higher feels smoother.
Dot pitch	Distance between same-coloured dots in a CRT. A smaller pitch gives a sharper picture.
Aspect ratio	Width to height, for example 16:9.

Graphics workstations

Definition: Graphics workstation

A graphics workstation is a high-performance computer system built for graphics-heavy professional work. It combines a fast processor, large memory, a powerful graphics processor, one or more high-quality monitors and specialised input devices.

Feature	Ordinary PC	Graphics workstation
Purpose	General office and home use	CAD, animation, simulation, medical and scientific visualization
Graphics hardware	Basic or entry-level	Powerful professional-grade GPU
Memory and processor	Moderate	Large RAM, many CPU cores
Display	Single standard monitor	High-resolution, colour-accurate, often multiple monitors
Input devices	Keyboard and mouse	Plus tablets, 3D mice, dials, or tracking devices

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Feature	Ordinary PC	Graphics workstation
Cost	Lower	Higher

Real-world examples

- An animation studio artist uses a workstation with a pen tablet and a colour-accurate monitor.
- A mechanical engineer rotates a 3D engine model on a CAD workstation with two screens: one for the model and one for drawings.
- A radiologist views high-resolution scans on a medical-grade monitor.

8. Input Devices

How the user talks to the graphics system

Input devices send positions, choices, text and commands to the computer. In graphics, we also care about **what kind of value** the device gives.

Logical input functions (useful for advanced answers)

- **Locator:** gives a position (x, y).
- **Pick:** selects a displayed object.
- **Choice:** selects one option from a set (a menu item).
- **Valuator:** gives a single number (a slider value).
- **String:** gives text.
- **Stroke:** gives a sequence of positions (a drawn path).

One physical device can act as several logical devices. A mouse can locate, pick and choose.

Device	How it works	Typical use	Point to note
Keyboard	Key presses produce character codes. Cursor and function keys help control the display.	Text, labels, commands, shortcuts	Not good for positioning

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Device	How it works	Typical use	Point to note
Mouse	Movement (tracked mechanically or optically) moves the screen cursor. Buttons select.	GUI, drawing, selection	Gives relative movement, not absolute position
Trackball	A ball is rolled with the fingers or palm; the housing stays fixed.	Desk-space-limited work, kiosks	Like an upside-down mouse
Spaceball	Responds to push, pull and twist on a ball, giving six degrees of freedom.	Rotating and moving 3D objects in CAD	The ball barely moves; it senses force
Joystick	A lever moves the cursor in the direction pushed.	Games, simulators, positioning	Some types sense pressure rather than movement
Data glove	Sensors in a glove measure finger bending and hand position.	VR, gesture control, training	Gives hand shape and position in 3D
Digitizer (graphics tablet)	A stylus is moved over a flat surface; the device reports absolute coordinates.	Tracing drawings, digital art, CAD	Absolute positioning, high accuracy
Image scanner	Scans a picture or document and stores it as pixel data.	Digitizing drawings and photos	Gives an image, not shapes
Touch panel	Detects finger contact on the screen (optical, electrical or acoustic methods).	Phones, ATMs, kiosks	Direct pointing at screen items
Light pen	A pen-shaped sensor detects the light of the screen spot beneath it and notes the moment of the beam passing.	Selecting items on older CRT screens	Needs CRT-style scanning to work, so it is obsolete
Voice system	A microphone and speech recognition turn spoken words into commands.	Hands-free control, accessibility	Needs training or good recognition software

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Absolute vs relative positioning

A **tablet** and a **touch panel** give absolute positions: touch the top-left, the cursor goes top-left. A **mouse** and **trackball** give relative movement: they report how far you moved, not where you are.

Practical example: choosing an input device

- Tracing a paper floor plan into a CAD system: **digitizer**.
- Walking through a virtual building: **data glove** or a head-tracking headset.
- Rotating a 3D engine model smoothly in all directions: **spaceball**.
- Self-service ticket machine: **touch panel**.

9. Common Student Mistakes

Check yourself before the exam

Mistake	Correct idea
Saying resolution means screen size.	Resolution is the number of pixels. Screen size is the physical diagonal. A small phone can have more pixels than a large old monitor.
Mixing up persistence and refresh rate.	Persistence is a property of the phosphor (how long it glows). Refresh rate is a property of the system (how often it is redrawn).
Saying the frame buffer stores shapes.	It stores pixel values only.
Forgetting to divide by 8 in frame-buffer calculations.	Bits to bytes: divide by 8. Then 1 KB = 1024 bytes.
Describing random scan as "pixel by pixel".	Random scan draws line by line from a display file.
Thinking retrace draws something.	During retrace the beam is blanked (off).

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Mistake	Correct idea
Calling LCD an emissive display.	LCD is non-emissive. It controls backlight and makes no light itself.
Using shadow mask for random scan, beam penetration for raster.	Beam penetration is linked to random scan. Shadow mask is linked to raster scan.
Confusing computer graphics with image processing.	Graphics creates images from data. Image processing changes existing images.
Drawing a CRT diagram without a control grid or focusing system.	Include cathode, control grid, focusing, accelerating anode, deflection, phosphor screen, and label each one.
Writing "mouse gives absolute position".	A mouse gives relative movement.

10. Bloom's Taxonomy Map

What level of thinking each concept needs

Level	Concept in this unit	Sample task
Remember	Definitions: pixel, frame buffer, persistence, refresh rate, retrace	Define persistence.
Understand	Working of CRT, raster scan, random scan, flat panels	Explain how a CRT creates a spot of light.
Apply	Frame-buffer size, number of colours, time per pixel	Calculate the memory needed for 1280 × 720 at 24 bpp.
Analyse	Raster vs random; shadow mask vs beam penetration; CRT vs LCD	Compare raster and random scan on memory, quality and use.
Evaluate	Choosing display and input devices for a need	Justify which display suits a wireframe plotter versus a game.
Create	Designing a graphics system or workstation	Propose a workstation setup for a small animation studio and justify each part.

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11. Interview Questions

With short model answers

1. What is a frame buffer and why is it needed?

It is memory that stores the colour of every pixel. The video controller reads it continuously to refresh the screen, so the picture stays visible and the CPU can work on something else.

2. Why does a CRT need refreshing, but an LCD pixel does not "fade"?

CRT phosphor glows only briefly after being hit, so the beam must redraw it. An LCD pixel holds its state by voltage. The panel is still refreshed at a fixed rate, but there is no glow decay to fight.

3. Why is raster scan more popular than random scan?

Raster handles filled areas, shading, photographs and millions of colours, and its refresh time does not depend on picture complexity. Memory became cheap, which removed random scan's memory advantage.

4. What is aliasing and how can it be reduced?

Jagged edges on slanted lines due to the pixel grid. Anti-aliasing blends edge pixels with the background, or the resolution can be increased.

5. What is the difference between resolution and screen size?

Resolution is the pixel count (for example 1920 × 1080). Screen size is the physical diagonal in inches. The same resolution on a smaller screen gives a sharper image.

6. What is the difference between a bitmap and a pixmap?

A bitmap uses 1 bit per pixel (two colours). A pixmap uses multiple bits per pixel (grey levels or colours).

7. What does a GPU do that a CPU does not do efficiently?

A GPU has many simple cores that perform the same operation on many pixels or vertices at once, which suits graphics. A CPU has fewer, more flexible cores.

8. What is interlacing?

Refreshing odd lines first and then even lines, in two fields per frame, to reduce flicker at low frame rates.

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9. Name an input device that gives six degrees of freedom.

> A spaceball (and, in VR, tracked controllers or gloves). It senses push, pull and twist in three directions each.

10. A client wants a display for a wiring-diagram plotter and another for a racing game. What would you choose?

> A line-drawing system (random-scan style or a vector plotter) is a natural fit for wiring diagrams. For the game, a raster display with a high refresh rate and strong GPU, because it needs colour, shading and smooth motion.

12. University Examination Questions

Typical patterns for this unit

These are typical question patterns for this topic. Check your university's past papers for the exact style and marks.

2-mark questions

1. Define computer graphics.
2. What is persistence of phosphor?
3. Define refresh rate and resolution.
4. What is a frame buffer?
5. Differentiate between bitmap and pixmap.
6. What is horizontal retrace?

5-mark questions

1. List any five application areas of computer graphics with examples.
2. Explain the working of a shadow-mask CRT.
3. Write a short note on flat-panel displays (emissive and non-emissive).
4. Differentiate between raster-scan and random-scan systems (any five points).
5. Explain any four input devices used in graphics.
6. Write a note on the direct-view storage tube.

10-mark questions

1. Explain the architecture of a raster-scan system with a neat diagram.

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2. Describe a refresh CRT with a diagram and explain persistence, resolution and aspect ratio.
3. Explain random-scan systems and compare them with raster-scan systems.
4. Describe the components of a graphics system. Explain the roles of the display processor and the video controller.
5. Discuss graphics monitors and workstations, and the input devices used with them.

13. 5-Minute Classroom Activity

"Pick the right hardware"

Setup (works in pairs)

Each pair gets one scenario. They choose **(a) raster or random scan, (b) a display technology, and (c) one input device**, and give one reason for each.

Scenario	Teacher's expected answer (do not show first)
A. A hospital wants to show high-detail X-ray images to doctors.	Raster; high-resolution, colour-accurate flat panel; mouse or touch for zoom.
B. A school wants a display for a pen plotter that draws circuit line diagrams.	Random-scan style line drawing; line-oriented display; digitizer or keyboard.
C. A game studio is testing a fast racing game.	Raster; high refresh rate LCD or OLED; game controller or joystick.
D. A railway ticket machine for first-time users.	Raster; rugged flat panel; touch panel.
E. A designer needs to rotate a 3D machine part freely.	Raster; large flat panel; spaceball with a mouse.

Time plan

- **Minute 1:** teacher reads the scenarios and assigns one per pair.
- **Minutes 2 and 3:** pairs discuss and write answers with reasons.
- **Minutes 4 and 5:** three pairs share. The class votes agree or disagree and the teacher corrects.

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Bonus (if time remains)

Calculate the frame-buffer size for scenario C at 1920 × 1080 with 24 bits per pixel.
Answer: 1920 × 1080 × 3 = 6,220,800 bytes, about 5.93 MB.

14. Short Recap

One minute before you leave

Ten things to remember

1. Computer graphics turns data into pictures (and supports interaction).
2. Major areas: CAD, presentation graphics, art, entertainment, education, visualization, GUI, VR and GIS.
3. A graphics system has input devices, CPU, memory, display processor/GPU, frame buffer, video controller and monitor.
4. A CRT has a cathode, control grid, focusing system, accelerating anode, deflection system and phosphor screen.
5. Persistence = glow time of phosphor. Refresh rate = redraws per second.
6. **Raster scan** sweeps every line and stores pixels in a frame buffer. **Random scan** draws only lines from a display file.
7. Shadow mask gives many colours (raster). Beam penetration gives few (random scan).
8. Flat panels are either **emissive** (plasma, LED/OLED) or **non-emissive** (LCD).
9. Frame buffer size = width × height × bits per pixel ÷ 8 (bytes).
10. Tablets and touch panels give absolute position. Mice and trackballs give relative movement.

15. Practice Set

Test yourself, then check the answers

A. Five MCQs

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Q1. Which display type stores the picture as intensity values for every screen pixel?

(a) Random-scan (b) Raster-scan (c) Pen plotter (d) Light pen

Q2. Persistence is the time taken for the phosphor glow to decay to what fraction of its original intensity?

(a) one-half (b) one-quarter (c) one-tenth (d) one-hundredth

Q3. In a shadow-mask colour CRT, each pixel position is made up of:

(a) one white phosphor dot (b) two phosphor dots (c) a triad of red, green and blue dots (d) four phosphor dots

Q4. The frame-buffer size for an 800 × 600 display with 16 bits per pixel is:

(a) 480,000 bytes (b) 960,000 bytes (c) 1,920,000 bytes (d) 7,680,000 bytes

Q5. Which of these is a non-emissive display?

(a) Plasma panel (b) LED display (c) LCD (d) Refresh CRT

Answers: Q1: (b). Q2: (c). Q3: (c). Q4: (b), since 800 × 600 = 480,000 pixels × 2 bytes = 960,000 bytes. Q5: (c).

B. Five short-answer questions

SA1. Differentiate between persistence and refresh rate.

Persistence is how long the phosphor keeps glowing after the beam leaves (a material property, measured as the decay to one-tenth intensity). Refresh rate is how many times per second the screen picture is redrawn (a system property). Short-persistence phosphor needs a higher refresh rate.

SA2. What is a frame buffer? Find its size for a 1280 × 720 display at 24 bits per pixel.

A frame buffer is memory holding the colour value of every screen pixel. Size = 1280 × 720 × 3 bytes = 921,600 × 3 = 2,764,800 bytes ≈ 2.64 MB.

SA3. Why is interlacing used?

When the full-frame refresh rate is low, flicker is visible. Interlacing refreshes odd lines first and even lines next, so the screen is partly updated twice as often, which reduces perceived flicker.

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SA4. Write any four differences between raster scan and random scan.

(1) Raster sweeps all lines; random scan draws only the picture lines. (2) Raster stores pixels in a frame buffer; random scan stores commands in a display file. (3) Raster refresh time is constant; random scan depends on the number of lines. (4) Raster supports shading and many colours; random scan has smooth lines but poor shading and limited colour.

SA5. Name four application areas of computer graphics with one example each.

CAD (designing a machine part in SolidWorks), presentation graphics (charts in Excel), entertainment (animated films and games), education and training (flight simulator). Others: visualization, GUI, VR and GIS.

C. Three long-answer questions

LA1. (10 marks) Explain the construction and working of a refresh CRT with a neat diagram. Define persistence, resolution and aspect ratio.

Answer outline:

- Diagram like Figure 3 with all parts labelled.
- Step-wise working: cathode, control grid, focusing, accelerating anode, deflection, phosphor glow, need for refresh.
- Definitions of persistence, refresh rate, resolution, aspect ratio.
- One line linking short persistence to high refresh rate.

LA2. (10 marks) Explain raster-scan and random-scan systems with architecture diagrams. Compare them in a table.

Answer outline:

- Definition and working of raster scan with Figure 5 (left) and Figure 6; roles of frame buffer, video controller and display processor; interlacing and retrace in brief.
- Definition and working of random scan with Figure 5 (right) and Figure 7; display file and display processor; flicker with many lines.
- Comparison table with at least six points (see Section 6), and a closing line on current use.

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LA3. (10 marks) Describe five input devices used in computer graphics. For each, explain how it works and give one application.

Answer outline: choose five from keyboard, mouse, trackball, spaceball, joystick, data glove, digitizer, scanner, touch panel, light pen and voice system. For each, write two lines of working and one application, and mention absolute vs relative positioning where relevant. Finish with the logical input functions (locator, pick, choice, valuator, string, stroke) for a stronger answer.

End of Unit 1 notes. For PDF: use your browser's Print, choose "Save as PDF", and switch on "Background graphics" to keep the colours.

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